

Tarea 2 Vectores

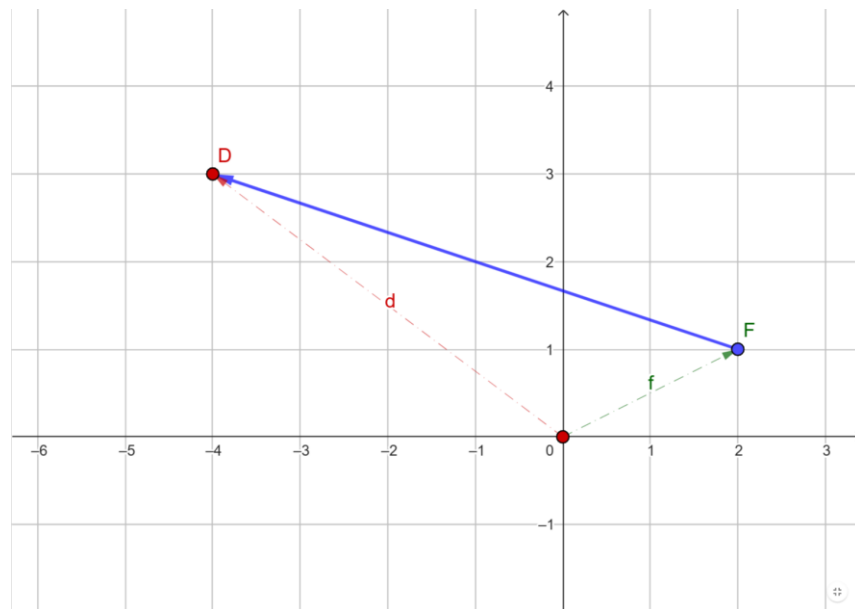
MATEMÁTICAS Y ESTADÍSTICA PARA INTELIGENCIA ARTIFICIAL
CARLOS RAMÍREZ MARTÍNEZ

Vectors (fundamentals)

I - Position Vectors 2D

Using the applet below

- a) Move F so that it has position Vector (2,1).
- b) Move D so that it has position vector (-4,3).
- c) Place the ends of the blue vector so that it becomes vector FD:



- d) State the components of FD.

$$f = OF = (2, 1)$$

$$d = OD = (-4, 3)$$

$$fd = FD = (-4 - 2, 3 - 1) = (-6, 2)$$

- e) State FD in terms of vectors f and d.

$$fd = FD = d - f$$

$$f-d = -2+4, 1-3, 2, -2$$

II- Vector Journeys 2D

Using the applet below

$u_1 = u$ and $v_1 = v$, move u_1 and v_1 about to help you visualise and answer the questions below. In terms of u and v state:

a) $AB = u$

b) $BC = v$

c) $AD = v$

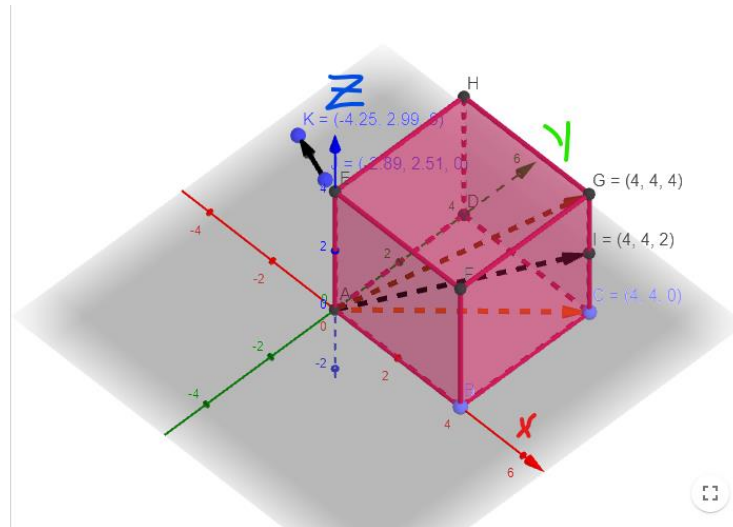
d) $CD = -u$

e) $AC = u + v$

f) $DB = u - v$

III- Position Vectors 3D

Using the applet above



- 1) List the coordinates of Points A,B,C,D,E,F,G,H,I. Notice that point A and O are the same.

$$O = A = (0, 0, 0)$$

$$B = (4, 0, 0)$$

$$C = (4, 4, 0)$$

$$D = (0, 4, 0)$$

$$E = (0, 0, 4)$$

$$F = (4, 0, 4)$$

$$G = (4, 4, 4)$$

$$H = (0, 4, 4)$$

$$I = (4, 4, 2)$$

- 2) State:

- a) the components of $\vec{OH} = \vec{AH} = \vec{h}$

$$\underline{OH = AH = (0 - 0, 4 - 0, 4 - 0) = (0, 4, 4)}$$

- b) the components of $\vec{AC}$

$$\underline{AC = (4 - 0, 4 - 0, 0 - 0) = (4, 4, 0)}$$

- c) the vector $\vec{BG}$ in terms of vectors $\vec{OB}$ and $\vec{OG}$.

$$OB = (4 - 0, 0 - 0, 0 - 0) = (4, 0, 0)$$

$$OG = (4 - 0, 4 - 0, 4 - 0) = (4, 4, 4)$$

$$\underline{BG = (4 - 4, 4 - 0, 4 - 0) = (0, 4, 4)}$$

d) the components of $\vec{EI}$

$$\underline{EI = (4 - 0, 4 - 0, 2 - 4) = (4, 4, -2)}$$

e) the vector $\vec{IE}$ in terms of $\vec{EI}$

$$\underline{IE = (0 - 4, 0 - 4, 4 - 2) = (-4, -4, 2)}$$

f) the components of $\vec{JK}$

$$\underline{JK = (-4.25 - (-2.89), 2.99 - 2.51, 0 - 0) = (-1.36, 0.48, 0)}$$